



PolyCrit: An Online Collaborative Platform for Polymer Characterization

Brinton King Eldridge¹, Dillon T.A. Baker¹, Yongmei Wang^{1,*}

Department of Chemistry, University of Memphis, Memphis, TN, 38152, USA

ARTICLE INFO

Keywords:

Liquid chromatography at critical conditions (LCCC)

Polymer

Database

Collaborative platform

PolyCrit

Machine learning

ABSTRACT

Polymer liquid chromatography at critical conditions (LCCC) is a chromatographic separation condition achieved by carefully balancing the interaction of a polymer with stationary and mobile phases to make the elution time of a polymer in chromatography independent of its molecular weight. By removing the dependence of elution time on polymer molecular weight, the LCCC then allows separation of polymer samples on the basis of secondary differences, such as topology, branching, and end-group functionality, that are otherwise difficult to resolve. Despite its potential, LCCC remains under-employed due to the complexity of its optimization and the scattered nature of existing data. To address these challenges, we developed *PolyCrit*, a database that organizes 428 critical chromatography conditions (characterized by 33 parameters) into a searchable and accessible online platform. *PolyCrit* centralizes decades of literature, providing detailed information on polymers, solvents, stationary phases, and chromatographic parameters. It features a quality scoring system to ensure data reliability and supports contributions from the research community through a validation process. By curating experimental critical conditions, *PolyCrit* reduces the need for extensive literature searches to utilize the powerful chromatographic technique. Additionally, *PolyCrit* invites current researchers to contribute to the database by submitting their own work. It can be found at <https://lccc.ywangcomp.org>.

1. Introduction

Synthetic polymer samples are seldom composed of identical copies of the same molecule; instead, they consist of a distribution of species that vary across multiple characteristics, such as molecular weight (M_w), chemical composition, functionality, chain architecture, and topology. Accurately and comprehensively characterizing these complex polymer systems' distributions is of significant importance to many experimental chemists, industrial researchers, and material scientists. One such umbrella of methodologies for characterization is known as liquid chromatography, and it is a versatile analytical technique essential for separating and analyzing components within a mixture based on their physical or chemical properties [1]. For example, a solution of polymers [2] varying in degrees of polymerization (a property that is proportional to M_w) can be separated by M_w using a method known as Size Exclusion Chromatography (SEC). In SEC, the stationary and mobile phases are chosen to minimize interactions with polymers, preventing their adsorption and ensuring that separation occurs solely based on differences in polymer's hydrodynamic volumes. SEC is commonly used to obtain molecular weight distribution in polymer samples. However, SEC cannot be used to obtain other molecular characteristics' distributions such as functionality, chemical composition, and architecture.

In response to the needs of characterizing these complex polymer systems, analytical polymer chemists have developed Interactive Chromatography (IC) methods that tap into the chemical specific interaction between polymers and column substrate. The collective IC systems include solvent gradient interactive chromatography (SGIC), thermal gradient interactive chromatography (TGIC), Liquid Adsorption Chromatography (LAC), etc. Unlike SEC, where polymer adsorption is minimized, IC systems intentionally allow polymers to adsorb onto the chromatography columns. The adsorption strength is carefully adjusted by modifying the solvent composition or temperature, which in turn enables detailed probing of the polymers' chemical composition and functionality. The fine-tuning of polymer adsorption on column substrates relies on one important concept in polymer physics, the critical adsorption point (CAP), which marks the condition that a given polymer is either adsorbed on the column surface or repelled from the column surface. The existence of CAP leads to a special type of polymer chromatography mode, namely, liquid chromatography at the critical condition (LCCC). At the LCCC condition, a polymer will elute independent of its molecular weight. In plots of elution time versus M_w , this behavior is distinctly represented by a vertical line (Fig. 1). At this point, the species' interaction with the stationary phase balances its interaction with the mobile phase (refer to Section 2 for more information). This balance enables separation based on subtle differences in

* Corresponding author.

E-mail addresses: bldrdge1@memphis.edu (B.K. Eldridge), dtbaker@memphis.edu (D.T.A. Baker), ywang@memphis.edu (Y. Wang).

<https://doi.org/10.1016/j.chroma.2025.465821>

Received 13 January 2025; Received in revised form 23 February 2025; Accepted 24 February 2025

Available online 7 March 2025

0021-9673/© 2025 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

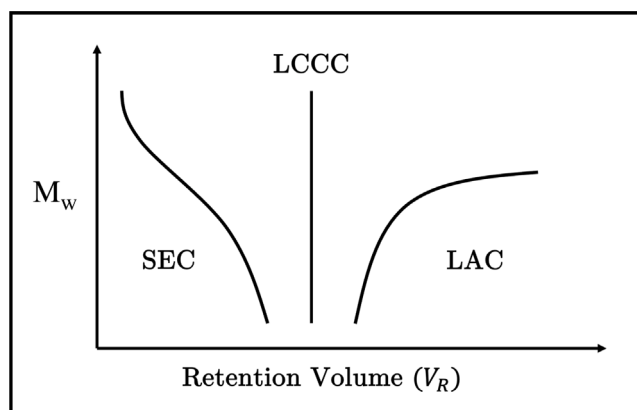


Fig. 1. Illustration of the three elution modes in polymer liquid chromatography: Size Exclusion Chromatography (SEC), Liquid Adsorption Chromatography (LAC), and Liquid Chromatography at the Critical condition (LCCC).

chemical or physical structure rather than M_w , making LCCC indispensable for characterizing complex polymer systems. LCCC is particularly valuable for analyzing a wide range of polymer characteristics that are challenging to distinguish using other chromatographic methods.

Numerous dedicated researchers have explored and identified critical conditions for polymers relevant to their fields. However, these investigations are dispersed across a wide range of chromatographic journals, resulting in a fragmented and scattered body of literature. Since its introduction in the late 1980s, LCCC has been referred to by various names and initialisms, further contributing to the lack of cohesion in the field: LC-CC, polymer critical condition (polymer CC), critical polymer chromatography (CPC), critical solvent condition (CSC), critical adsorption point (CAP, CPA, LC CAP), liquid adsorption critical chromatography (LACC), liquid chromatography at critical conditions of adsorption (LCCCA, LACCC), and liquid chromatography of polymers at the exclusion-adsorption transition point (LC EATP, LC PEAT) [1]. Though each name emphasizes a specific aspect of LCCC, they nevertheless widen the search terms necessary for research.

Being that each addition to the field is reported individually, as the field grows, the depth at which a researcher must search to find specific conditions also grows. Currently, a researcher would have to begin with an extensive literature review to find the critical conditions relevant or similar to their polymer of choice. However, the scattered nature of existing LCCC data poses significant challenges to researchers seeking to adopt or optimize this technique. No unifying resource exists to catalog and report all LCCC findings that are publicly available, and such absences are being felt throughout the polymer field [3]. Without a unifying resource, researchers would have to work with fragmented information, making the process of finding LCCC conditions a far more daunting task.

Additionally, the critical condition identified in LCCC contains rich physiochemical information about polymers, solvents, and stationary phases. Since this field is fundamentally data-centric, a collection of LCCC data can be further mined to reveal the underlying principles/properties that may not be easily observed. Without a dedicated online platform to aggregate these conditions, the integration of machine learning into LCCC conditions would remain severely limited by the lack of accessible and comprehensive data.

To address this challenge and others that surround it, we have developed a centralized, dynamic online platform with a comprehensive database of polymer liquid critical chromatography critical conditions titled *PolyCrit*. This website organizes and currently reports 428 critical conditions in its initial release; offers a user-friendly, searchable interface for ease-of-access; and welcomes current researchers to add to the database to increase the breadth of data it contains.

2. Background

Liquid chromatography encompasses a wide range of methods, including Size Exclusion Chromatography (SEC) and Liquid Adsorptive Chromatography (LAC), each distinguished by its separation mechanism. SEC separates molecules primarily based on size, utilizing a porous stationary phase where smaller molecules are retained longer due to their ability to enter the pores. This method is particularly effective for determining M_w distributions of polymers and proteins. In contrast, LAC separates molecules based on chemical interactions with the stationary phase, such as hydrophobic, polar, or ionic interactions. This makes LAC well-suited for analyzing chemical properties and structural nuances in complex mixtures.

While SEC and LAC are widely used, their limitations become evident when characterizing polymers. Polymers, due to their high range of M_w 's and propensities for chemical interaction, often require hybrid or alternative approaches for effective separation. Two-dimensional liquid chromatography (2D-LC) and solvent/temperature-gradient interaction chromatography (S/TGIC) are examples of advanced methods that address these challenges [4].

However, LCCC offers a specialized solution by enabling polymer separations independent of M_w under critical conditions. This capability allows researchers to distinguish between polymers with similar M_w but differing structures. For instance, LCCC can separate linear polymers from branched or cross-linked polymers, even when these species have similar M_w . LCCC's sensitivity to structural differences extends its utility to analyzing branching degrees and other architectural features of polymers. Moreover, removing one of the most significant confounding variables in separations, M_w , provides researchers with a practical, hands-on approach to prepare highly pure polymer solutions for secondary separations, eliminating the need for industrial equipment or manufacturing controls [5].

The theory of LCCC relies on achieving a "critical condition" where the elution of a polymer is unaffected by its M_w but depends on its chemical interaction with the stationary phase of column substrates in chromatographic system. A full elaboration of the theory behind LCCC can be found in a previous work of ours [6], but we give a brief summary of the necessary terminology and equations here.

The Critical Adsorption Point (CAP) in liquid chromatography represents a pivotal transition in how polymer chains interact with a solid surface or a stationary phase. When polymer chains approach a surface, their behavior depends on a balance between entropic and enthalpic effects. In the absence of attractive interactions (enthalpic gains), the chains lose configurational entropy as they are restricted near the surface, leading to what is known as a depletion layer, where polymer density near the surface is lower than in the bulk phase.

As the strength of polymer-surface interaction increases, there comes a critical interaction strength where the entropy loss is exactly balanced by the enthalpic gain. At this point, the polymer density profile near the surface becomes flat, meaning there is neither depletion nor excess density. This critical interaction strength defines the CAP. Beyond the CAP, when the enthalpic gain outweighs the entropy loss, polymer chains begin to adsorb preferentially onto the surface, leading to an excess density layer, where the concentration of polymers near the surface exceeds that in the bulk.

The retention volume V_R of polymers in chromatography is related to the partition coefficient K by $V_R = V_0 + KV_p$, where V_0 is the void (or mobile phase) volume and V_p is the stationary phase's pore volume, respectively [7]. Using the Gaussian chain model, one can approximate the value for K [4],

$$K = 1 + \frac{2R}{D} \left[\frac{\exp((cR)^2) [1 - \text{erf}(-cR)] - 1}{cR} - \frac{2}{\sqrt{\pi}} \right] \quad (1)$$

where $\text{erf}(x)$ is the error function,

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x \exp(-t^2) dt. \quad (2)$$

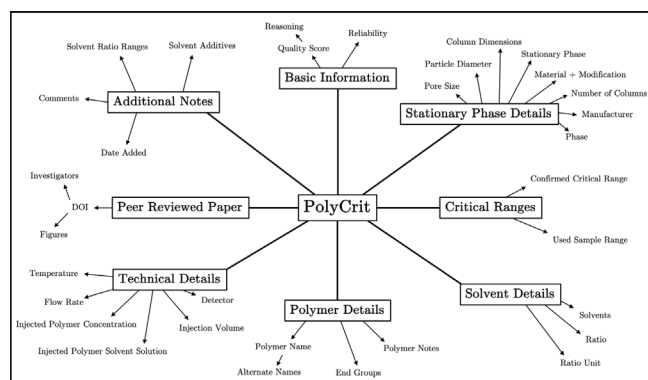


Fig. 2. Cluster diagram detailing different tracked parameters and their respective categories. Arrows between parameters indicates flow of creation.

The partition coefficient K , influenced by the ratio of the polymer's radius of gyration (R , a measure of M_w) to the pore size (D), determines how polymers interact with the stationary phase. In Eq. (1), c is the adsorption interaction parameter used to measure the potential interaction energy between the polymer and the stationary phase. In SEC, $c < 0$, indicating that the interaction energy of the polymer is weaker than the critical threshold, resulting in elution based primarily on molecular weight (M_w). Conversely, in LAC, $c > 0$, signifying stronger polymer interactions with the stationary phase, leading to adsorption. At the CAP, $c = 0$ and thus K approaches 1, independent of pore size or chain length, reflecting a balance where adsorption neither dominates nor depletes polymer density near the wall [8]. However, in practice, seeking conditions for $K = 1$ has been shown to not be particularly viable in identifying LCCC [6,9–11]. Instead, experimentalists typically will simply identify the phenomenon based on its characteristic vertical line (as in Fig. 1).

3. Materials and methods

In developing this database, we undertook an extensive literature review to compile LCCC data spanning several decades. This review included studies from as early as 1976 up to the most recent publications in 2023, reflecting the evolution and refinement of LCCC methodologies over time. Particularly, we searched through major chromatographic journals picking papers mentioning relevant key terms also found in this paper. Review papers written by Macko et al. [12] in 2003 and Baumgaertel et al. [13] in 2012, as well as an early database paper by Trathnigg et al. [14] in 2007 were of great help during this review. Their works inspired the creation of this database platform, and their precedents helped us decide what experimental conditions needed to be tracked.

The selection process was rigorous; we focused on studies that provided detailed chromatographic conditions, including polymer type, solvent combinations, stationary phases, and other relevant parameters (a full list can be found in Fig. 2). Each paper was carefully examined to verify that critical conditions were accurately achieved and documented. In total, we reviewed 312 papers on the topic, recorded 625 conditions, and selected 428 unique LCCC conditions, each validated and curated for inclusion in the database.

3.1. Curation

Some entries were initially reported by other publications as being critical but upon further review did not present sufficient evidence and were removed from the dataset. Additionally, similar systems were present in the literature multiple times. Such entries were combined at the discretion of the reviewer into one entry. If some conditions were

Table 1

An example of data curation by consolidating similar entries where the top and bottom tables are examples of the before and after.

Polymer	Solvent	Solvent ratio
Polystyrene	DCM, ACN	58.0, 42.0
Polystyrene	DCM, ACN	58.2, 41.8

Polymer	Solvent	Solvent ratio	Solvent ratio ranges
Polystyrene	DCM, ACN	58.1, 41.9	58.0–58.2% DCM

Table 2

Reliability scores for assessing the independence of molecular weight (M_w) and retention time in critical condition data.

Reliability	Reasoning
Excellent	5 or more data points clearly indicating an independent (vertical) relationship between M_w and retention time.
Good	4 or more data points roughly indicating an independent relationship, but not obviously vertical.
Low	2 to 3 data points roughly indicating an independent relationship, but not obviously vertical.
Indirect	Critical conditions stated, but no evidence given.

similar enough to be combined but not exact, a note was left to describe the range of the critical conditions (see Table 1 for an example).

To combine two entries, nearly all conditions had to match exactly, with allowances for differences in only one or two criteria. Uncombined records were retained for documentation purposes. After this thorough selection process, we turned our attention to assessing the quality of each condition, recognizing that the reliability of the data is essential for meaningful analysis and comparison.

3.2. Solvent ratio unit

Several critical conditions involved solvent ratios expressed in units of wt% or mol%. Given the potential for machine learning applications in this project, we decided to convert all these ratios to vol%, the most commonly used unit among the reviewed studies.

Solvent ratios were normalized to ensure consistency. This normalization step adjusts the ratios so that their sum equals 100%, calculated as:

$$\text{Normalized Ratio}_i = \frac{\text{Original Ratio}_i}{\sum_j \text{Original Ratio}_j} \times 100\% \quad (3)$$

where Original Ratio_i represents the proportion of the i th solvent in the mixture, and the denominator is the total of all solvent ratio components in that entry.

Next, solvent ratios were converted to volume units (vol%) for uniform representation. For ratios initially provided in weight (wt%), the conversion utilized the density (d) of each solvent:

$$\text{Volume Ratio}_i = \frac{\text{Weight Ratio}_i}{d_i} \quad (4)$$

For ratios provided in moles (mol%), the conversion accounted for both the molecular weight (M_w) and density (d) of each solvent:

$$\text{Volume Ratio}_i = \frac{\text{Mole Ratio}_i \cdot M_{w,i}}{d_i} \quad (5)$$

Finally, the ratios were again normalized to 100%. These calculations ensured that all solvent ratios were expressed in a comparable volumetric format.

3.3. Quality rating

By definition, the retention time and the M_w should have an independent relationship at LCCC (the vertical line shown in Fig. 1). However, since a continuous range of polymer weights cannot be evaluated, a relatively small number of vertically aligned data points

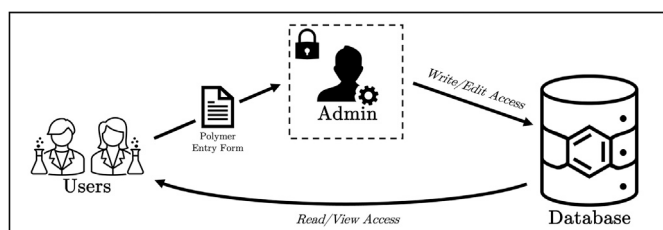


Fig. 3. General outline for Read and Write access to the database.

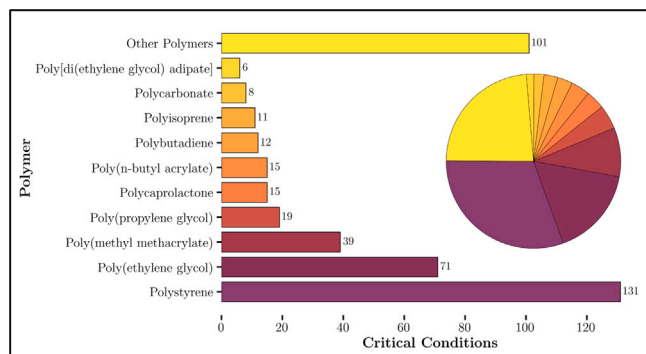


Fig. 4. Ten most common polymers present in the *PolyCrit* database.

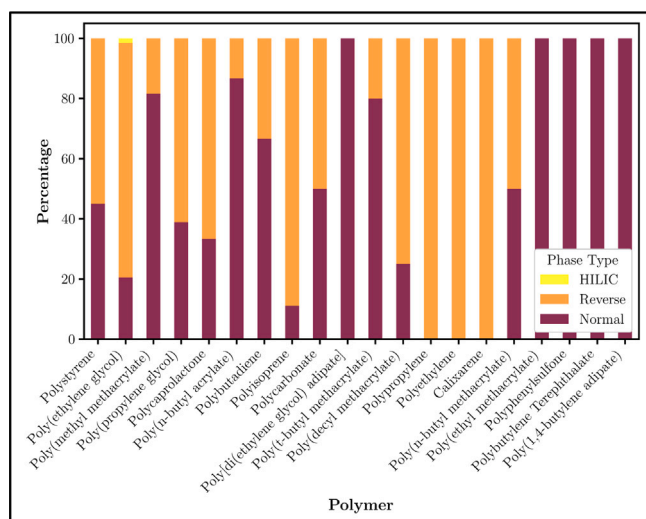


Fig. 5. Relative percentages of stationary phase type for the twenty most common polymers in *PolyCrit*.

are sufficient to prove a set of conditions as critical. Consequently, more data points provided for an experiment should be reflected in the data's reliability; thus, each critical condition was graded with a Reliability score (Table 2).

Nonetheless, various other factors may adversely affect the data, so each entry is assigned an overall Quality score to account for these influences. The Quality score starts at ten and loses a point for every negative quality present in an entry:

- indirect Reliability score (special reduction: −2)
- low Reliability score
- missing “Confirmed Critical Range”
- missing “Particle Diameter” of stationary phase
- missing “Pore Size” of stationary phase
- missing “Column Dimensions” of stationary phase

- unknown stationary phase “phase” (normal, reverse, etc.)
- unable to access paper (by our team at the point of this publication)
- had to be translated to English
- critical conditions present in only one paper

While a lower Quality rating does not imply any inaccuracy in the associated entry, it serves as a prompt for viewers to conduct further investigation. Each relevant entry includes at least one DOI hyperlink for convenient access to the resource, although some older resources or those from certain publication mediums may lack a DOI. To facilitate seamless access to these resources and support deeper investigation, we implemented a structured, user-friendly interface designed for researchers into the website.

3.4. Database access permissions

The website was designed to serve as a dynamic, user-friendly platform that researchers can easily navigate. To begin, all guests to the website can access and browse the database once they've signed in through email and become users, allowing interested parties to explore the curated data (Fig. 3). Additionally, this process allows us to keep track of members of the field who are utilizing our database. The website's structure is designed to be intuitive, enabling visitors to quickly find the information they need and to draw meaningful conclusions from the data. Additionally, the database is equipped with advanced search and filter options, making it possible to narrow down results based on specific criteria or combinations of conditions.

We acknowledge that some papers containing critical conditions were likely missed in this review, and we hope researchers will help us by reporting their work directly to the website. For such researchers wishing to contribute their own data to the database, they must contact a site administrator to register an account. In this process, the credentials of each contributing researcher will be verified, ensuring that all submissions come from reputable sources.

After verification, the researcher gains access to the polymer entry page, a dedicated area of the website where they can submit detailed information about their peer-reviewed LCCC conditions. Here, researchers are guided to enter specific parameters aligned with the standards outlined in Fig. 2, ensuring consistency and clarity across all submissions.

Finally, each new entry undergoes a review by an administrator before being added to the website's database. During this review, the administrator checks both the content of the submission and its accompanying peer-reviewed paper to confirm its validity.

4. Results and discussion

To better present the breadth of literature compiled in *PolyCrit*, we have produced several figures. Fig. 4 illustrates the distribution of unique critical conditions across the ten most common polymers in the *PolyCrit* database (though it currently contains 64 unique polymers). This figure highlights the diversity in critical condition data available for each polymer type, with Polystyrene having the largest number of unique conditions at 131, followed by Poly(ethylene glycol) with 71 conditions. In contrast, polymers like Polybutylene Terephthalate and Poly(ethyl methacrylate) have only a few critical conditions listed, which could indicate either limited application or a lack of comprehensive studies for these polymers. The diversity in the number of conditions across polymers highlights the value of *PolyCrit* as a consolidated resource, where underrepresented polymers can be more readily identified, potentially guiding future research efforts.

Fig. 5 provides a valuable overview of the distribution of stationary phase types — Normal, Reverse and Hydrophilic Interaction Chromatography (HILIC) — across the 20 most common polymers in the *PolyCrit* database. In Normal phases, the column substrate is polar

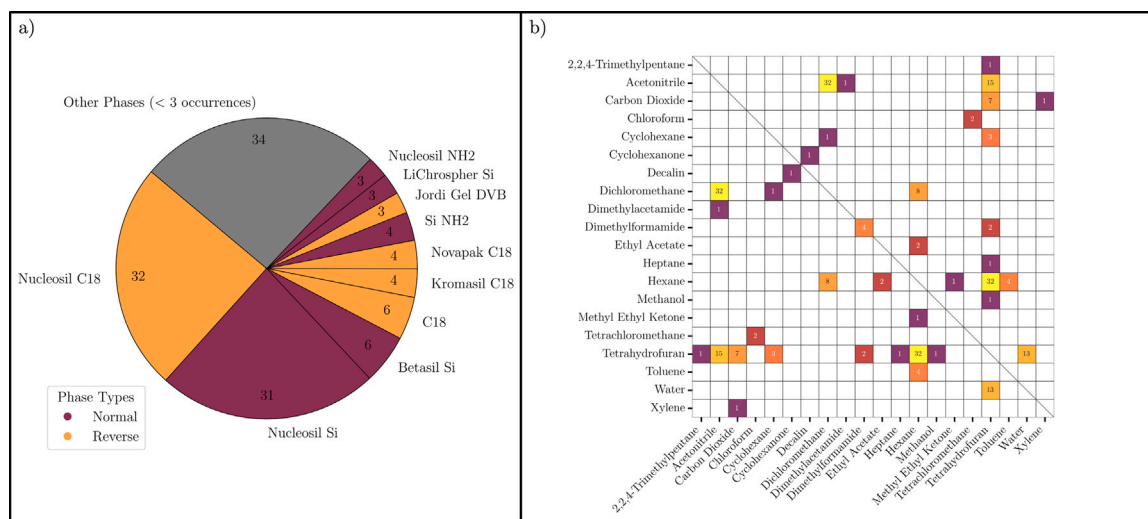


Fig. 6. (a) Pie chart of stationary phases reported used for Polystyrene, colored by Stationary Phase Type, and (b) heatmap of the solvent pairings used for Polystyrene. Numbers in each solvent pair indicate number of entries present in *PolyCrit*. Diagonal line highlights the mirrored quality of the heatmap.

Table 3

The five most common solvent solutions for Polystyrene entries with normal and reverse-phase columns. A (P) or (NP) indicates a solvent as either polar or nonpolar, respectively.

Normal phase	
Solvent solutions	Occurrences
Hexane (NP), THF (P)	35
MEK (P), Cyclohexane (NP)	10
DCM (P), Hexane (NP)	9
TCM (NP), Chloroform (NP)	9
THF (P), Cyclohexane (NP)	8
Reverse phase	
Solvent solutions	Occurrences
THF (P), Water (P)	38
DCM (P), ACN (P)	33
ACN (P), THF (P)	23
Acetone (P), Water (P)	19
Water (P), ACN (P)	17

— typically porous silica beads — whereas in Reverse phases, the substrate is hydrophobic, with C18-terminated silica being a common example. The choice of solvent is closely tied to the phase type, users are encouraged to visit the website to explore specific solvent pairings captured in the database, which can aid in selecting the most commonly used or effective solvents for their desired stationary phase type and polymer.

To give a sample of the critical conditions for the most common polymer Polystyrene, Fig. 6 provides both the stationary phases (Fig. 6a) and the solvent pairs (Fig. 6b). The large proportion of Nucleosil Si/C18 stationary phases is a commonality among many of the other critical conditions in *PolyCrit*. Notice in Fig. 6b the small portion of critical solutions that take a single solvent (Dimethylformamide) instead of a pairing. Critical solutions containing three solvents are necessarily excluded from Fig. 6b by the nature of a heat map but can be found on *PolyCrit*.

Additionally, since as many as 33 parameters were tracked from each chromatographic entry, we are able to track the name, phase, and end group modifications of each stationary phase present in *PolyCrit*. In Table 3, we present the five most common solvent solutions used for the two main stationary phases (Normal and Reverse) of the columns. Our review of the data confirms that it is common practice to pair normal-phase columns with nonpolar solvents and reverse-phase columns with polar solvents—a standard approach in the field.

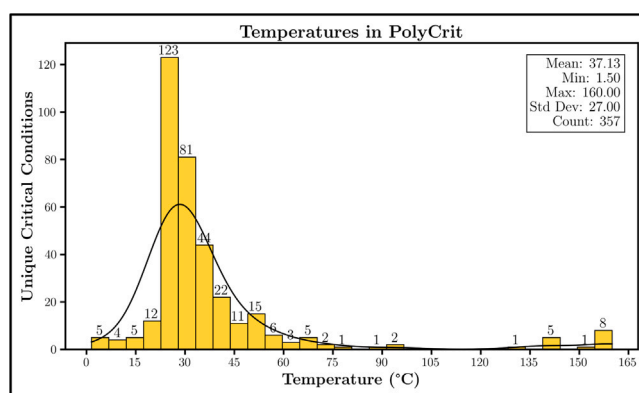


Fig. 7. Histogram of temperature with a box of key statistics. Black curves show the probability density function.

The last two figures (Figs. 7 and 8) display histograms of numeric-based parameters. Specifically, Fig. 7 gives an enlarged view (as compared to the smaller histograms of Fig. 8) of how temperature ranges across the critical conditions in *PolyCrit*. Note that the number of recorded temperature values (357) is different from the total entry count of 428; entries without a reported temperature have this parameter left blank. This approach applies to all parameters in the database; blank parameters were not replaced by default values. Naturally, the majority of temperatures lies close to room temperature (25 °C), but several studies can be found that were conducted with high temperature columns (> 100 °C).

In Fig. 8, all the numeric-based parameters are shown. Some histograms (marked with an asterisk) are sorted to only present the bottom 95th percentile of the data in order to remove high-valued outliers. However, such information is still preserved in the maximums recorded in the key statistics boxes.

Statistics presented in this section serve to give the reader a quick reference to information about the database, but many other comparisons and statistics can be made by the curious reader from the many entries present in *PolyCrit*.

5. Conclusion

The development of this database marks a significant advancement in the field of polymer research, with the potential to impact both

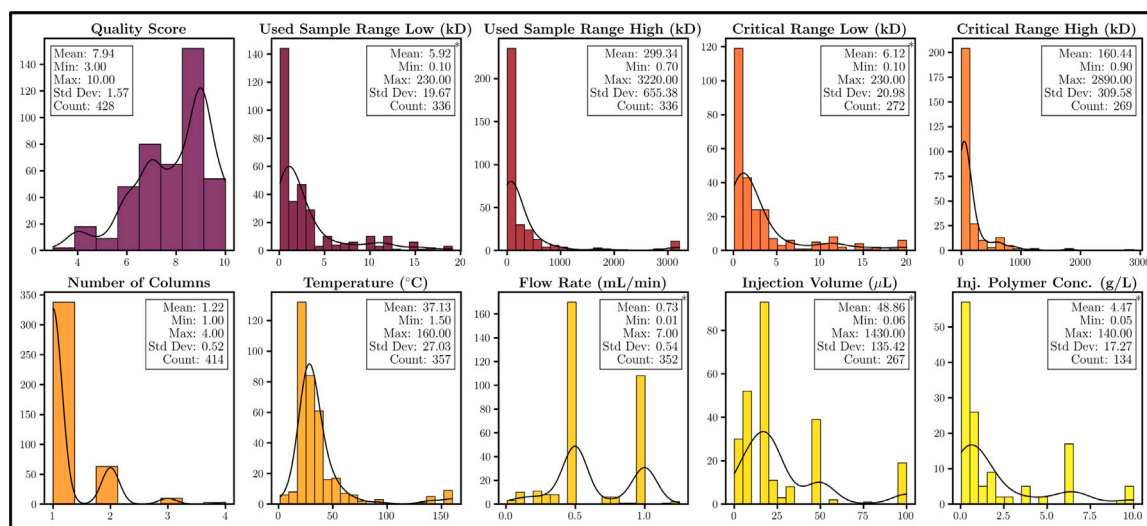


Fig. 8. Histograms of tracked numeric parameters with a box of key statistics. Vertical and horizontal axes display the count and value for each parameter. Black curves show the probability density function. Asterisks denote that the histogram displays only the lower 95th percentile.

current and future studies. By centralizing and streamlining access to critical chromatography conditions, this platform addresses long-standing challenges faced by researchers in accessing, sorting, and interpreting this information. Immediately, this resource offers polymer scientists a powerful tool for quickly identifying conditions that would otherwise require time-intensive literature searches. Over time, as the dataset expands and more researchers contribute their own findings, the platform will grow into a comprehensive resource, enabling advanced data analysis, cross-referencing, and ultimately accelerating the pace of discovery in the field. This database not only supports detailed studies of polymer properties but also opens up possibilities for exploring new critical conditions and optimizing experimental designs in an increasingly efficient manner.

Compared to existing resources, which often consist of static, printed compilations or small, standalone datasets, our database stands out as a dynamic, searchable, and continuously updated platform. It eliminates the need for repetitive data extraction from numerous studies by organizing conditions in a single, easily navigable repository. Unlike other resources, which tend to lose relevance over time, our platform is designed to grow with the field, incorporating both historical and newly submitted data. Additionally, as the field grows to include more critical data, we encourage researchers to use our platform to share their findings. The inclusion of submission capabilities empowers the community to participate actively in maintaining and expanding the dataset, fostering collaboration and reducing duplication of effort.

Beyond serving as a database, *PolyCrit* lays the groundwork for transformative applications in machine learning (ML) offering opportunities to uncover hidden patterns and predict critical conditions with greater accuracy. The current collection of polymers has already been converted to many descriptors we hope to use for ML purposes. One can elicit many numeric descriptors of a polymer including Hansen Solubility Parameters [15,16], as well as structural descriptors such as SMILES [17]. With sufficient data, ML algorithms could be trained to predict optimal critical chromatography conditions for new polymers based on structural or compositional similarities to previously characterized samples. This predictive capability would allow researchers to save time and resources by minimizing trial-and-error in laboratory experiments, offering initial conditions for experimentation that are likely to yield high-quality LCCC separations.

Furthermore, as the dataset grows, ML models could help identify underlying trends across different polymer types, such as correlations between branching patterns, M_w , and optimal separation conditions. Such insights could, in turn, guide the development of new polymer

formulations and analytical methods, further advancing the field of liquid chromatography and broadening the applications of LCCC. To that end, we are currently working on developing a ML model to predict features of polymer critical conditions using our curated database. In the model, we plan on emphasizing principles of green chemistry to transition the field to environmentally sustainable solvents. We hope to bring this to our online platform to bolster the website's helpfulness as a tool for researchers. By integrating the power of data science and ML with experimental chromatography, *PolyCrit* has the potential to revolutionize both the study and application of LCCC.

CRediT authorship contribution statement

Brinton King Eldridge: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Dillon T.A. Baker:** Writing – review & editing, Investigation, Data curation. **Yongmei Wang:** Writing – review & editing, Supervision, Investigation, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT to assist with rephrasing sentences and organizing key talking points. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

In preparation of this project, we would like to thank Nathan Dixon, Hunter Jackson, and Matthew Hosier, three University of Memphis Computer Science alums who helped build the initial framework for this website.

Appendix A. Supplemental information

As to give proper credit to the papers present in *PolyCrit*, we cite them now [14,18–232].

Data availability

PolyCrit can be found at <https://lccc.ywangcomp.org>.

References

- [1] T. Chang, Recent Advances in Liquid Chromatography Analysis of Synthetic Polymers, Springer Berlin Heidelberg, Berlin, Heidelberg, 2003, pp. 1–60.
- [2] T. Macko, R. Brüll, Y. Zhu, Y. Wang, A review on the development of liquid chromatography systems for polyolefins, *J. Sep. Sci.* 33 (22) (2010) 3446–3454.
- [3] D.J. Audus, J.J. de Pablo, Polymer informatics: opportunities and challenges, *ACS Macro Lett.* 6 (10) (2017) 1078–1082.
- [4] W. Radke, Polymer separations by liquid interaction chromatography: Principles – prospects – limitations, *J. Chromatogr. A* 1335 (2014) 62–79.
- [5] D. Berek, Coupled liquid chromatographic techniques for the separation of complex polymers, *Prog. Polym. Sci.* 25 (7) (2000) 873–908.
- [6] Y. Wang, Theory and Principles of Interaction Chromatography, ACS Symposium Series, pp. 19–30, ch. 2.
- [7] E.F. Casassa, Theoretical models for peak migration in gel permeation chromatography, *J. Phys. Chem.* 75 (26) (1971) 3929–3939.
- [8] A.A. Gorbunov, A.M. Skvortsov, Statistical properties of confined macromolecules, *Adv. Colloid Interface Sci.* 62 (1) (1995) 31–108.
- [9] Y. Gong, Y. Wang, Partitioning of polymers into pores near the critical adsorption point, *Macromol.* 35 (19) (2002) 7492–7498.
- [10] S. Orelli, W. Jiang, Y. Wang, A computational investigation of the critical condition used in the liquid chromatography of polymers, *Macromol.* 37 (26) (2004) 10073–10078.
- [11] Y. Wang, A. Masur, Y. Zhu, J. Ziebarth, Partitioning of star branched polymers into pores at three chromatography conditions, *J. Chromatogr. A* 1217 (39) (2010) 6102–6109.
- [12] T. Macko, D. Hunkeler, Liquid Chromatography under Critical and Limiting Conditions: A Survey of Experimental Systems for Synthetic Polymers, Springer Berlin Heidelberg, Berlin, Heidelberg, 2003, pp. 62–136.
- [13] A. Baumgaertel, E. Altıntaş, U.S. Schubert, Recent developments in the detailed characterization of polymers by multidimensional chromatography, *J. Chromatogr. A* 1240 (2012) 1–20.
- [14] B. Trathnigg, M.I. Malik, O. Jamelnik, N.V. Cuong, A data base for polymer chromatography: Dependence of interaction parameters on mobile phase composition, *Anal. Chim. Acta* 604 (1) (2007) 39–44.
- [15] S. Abbott, C.M. Hansen, Hansen Solubility Parameters in Practice, Hansen Solubility, 2008.
- [16] C.M. Hansen, Hansen Solubility Parameters: A User's Handbook, CRC Press, 2007.
- [17] S. Choi, J. Lee, J. Seo, S.W. Han, S.H. Lee, J.-H. Seo, J. Seok, Automated BigSMILES conversion workflow and dataset for homopolymeric macromolecules, *Sci. Data* 11 (1) (2024) 371.
- [18] P. Nefedov, T. Zhmakina, Effect of temperature, solvent composition and pressure on the adsorption and chromatography of polystyrenes on macroporous glasses, *Polym. Sci. U. S. S. R.* 23 (2) (1981) 304–312.
- [19] S.T. Balke, R.D. Patel, Orthogonal chromatography, 281–310, ch. 16.
- [20] A.V. Gorskov, V.V. Jevreinov, B. Lausecker, H. Pasch, H. Becker, G. Wagner, Flüssigchromatographie von makromolekülen unter kritischen Bedingungen. Zur auftrennung von poly(oxypropylenpolyolen) nach ihrer funktionalität, *Acta Polym.* 37 (11–12) (1986) 740–741.
- [21] A. Gorskov, T. Prudskova, V. Gur'yanova, V. Evreinov, Functional type separation of oligocarbonates by liquid chromatography under critical conditions: Effects of pore size, *Polym. Bull.* 15 (5) (1986).
- [22] V. Gur'yanova, A. Pavlov, Step-by-step analysis of polycondensation oligomers and polymers by liquid chromatography, *J. Chromatogr. A* 365 (1986) 197–204.
- [23] A. Gorskov, H. Much, H. Becker, H. Pasch, V. Evreinov, S. Entelis, Chromatographic investigations of macromolecules in the 'critical range' of liquid chromatography, *J. Chromatogr. A* 523 (1990) 91–102.
- [24] H. Krüger, H. Much, G. Schulz, G. Wehrstedt, Determination of functionality and molecular weight distributions in α -hydroxy- ω -benzyloxyoligo(1,3,6-trioxane)s using HPLC methods, *Die Makromol. Chem.* 191 (4) (1990) 907–920.
- [25] T. Zimina, Y. Kever, Y. Melenevskaya, V. Zgonnik, B. Belen'kii, Experimental verification of the concept of chromatographic 'invisibility' in critical chromatography of block copolymers, *Polym. Sci. U. S. S. R.* 33 (6) (1991) 1250–1254.
- [26] H. Pasch, C. Brinkmann, H. Much, U. Just, Chromatographic investigations of macromolecules in the 'critical range' of liquid chromatography, *J. Chromatogr. A* 623 (2) (1992) 315–322.
- [27] T. Zimina, J. Kever, E. Melenevskaya, A. Fell, Analysis of block copolymers by high-performance liquid chromatography under critical conditions, *J. Chromatogr. A* 593 (1–2) (1992) 233–241.
- [28] T.M. Zimina, A.F. Fell, J.B. Castledine, Diode-array detection with spectral suppression in size exclusion and adsorption chromatography for rapid evaluation of chemical composition in block copolymers of styrene and methyl methacrylate, *Polymer* 33 (19) (1992) 4129–4135.
- [29] H. Pasch, C. Brinkmann, Y. Gallot, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 4. Analysis of poly(styrene-*b*-methyl methacrylate), *Polymer* 34 (19) (1993) 4100–4104.
- [30] H. Pasch, M. Augenstein, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 5. Characterization of block copolymers of decyl and methyl methacrylate, *Die Makromol. Chem.* 194 (9) (1993) 2533–2541.
- [31] H. Pasch, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 3. Analysis of polymer blends, *Polymer* 34 (19) (1993) 4095–4099.
- [32] H. Pasch, Y. Gallot, B. Trathnigg, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 7. Analysis of the poly(methyl methacrylate) block in poly(styrene-block-methyl methacrylate), *Polymer* 34 (23) (1993) 4986–4989.
- [33] B. Trathnigg, D. Thamer, X. Yan, S. Kinugasa, Analysis of polyethers by isocratic HPLC with universal detectors. I. optimization of chromatographic conditions, *J. Liq. Chromatogr.* 16 (12) (1993) 2439–2452.
- [34] P.J.C.H. Cools, A.M. Van Herk, A.L. German, W. Staal, Critical retention behavior of homopolymers, *J. Liq. Chromatogr.* 17 (14–15) (1994) 3133–3143.
- [35] R.-P. Krüger, H. Much, G. Schulz, Determination of functionality and molar mass distribution of aliphatic polyesters by orthogonal liquid chromatography. I. Off-line investigation of poly(1,6-hexanediol adipates), *J. Liq. Chromatogr.* 17 (14–15) (1994) 3069–3090.
- [36] H. Pasch, M. Augenstein, B. Trathnigg, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 6. Analysis of the methyl methacrylate block in block copolymers of decyl and methyl methacrylate, *Macromol. Chem. Phys.* 195 (2) (1994) 743–750.
- [37] H. Pasch, I. Zammert, Chromatographic investigations of macromolecules in the critical range of liquid chromatography. VIII. Analysis of polyethylene oxides, *J. Liq. Chromatogr.* 17 (14–15) (1994) 3091–3108.
- [38] B. Trathnigg, D. Thamer, X. Yan, B. Maier, H.-R. Holzbauer, H. Much, Characterization of ethoxylated fatty alcohols using liquid chromatography with density and refractive index detection II. Quantification in liquid chromatography under critical conditions, *J. Chromatogr. A* 665 (1) (1994) 47–53.
- [39] P. Kitz, R.-P. Krüger, H. Much, G. Schulz, Two-dimensional chromatography for the deformation of complex copolymers, 1995, pp. 223–241.
- [40] H. Pasch, K. Rode, Use of matrix-assisted laser desorption/ionization mass spectrometry for molar mass-sensitive detection in liquid chromatography of polymers, *J. Chromatogr. A* 699 (1–2) (1995) 21–29.
- [41] V.B. Gancheva, N.G. Vladimirov, R.S. Velichkova, Chromatographic investigations of polyacetals in the critical range of liquid chromatography, 1. Analysis of 9-anthrylmethyl-terminated poly(1,3,6-trioxacyclooctane)s, *Macromol. Chem. Phys.* 197 (5) (1996) 1757–1770.
- [42] R. Krüger, H. Much, G. Schulz, O. Wachsen, New aspects of determination of polymer heterogeneity by 2-Dimensional orthogonal liquid chromatography and maldi-tof-MS, *Macromol. Symp.* 110 (1) (1996) 155–176.
- [43] R.P. Krüger, H. Much, G. Schulz, Determination of polymer heterogeneity by two-dimensional orthogonal liquid chromatography, *Int. J. Polym. Anal. Charact.* 2 (3) (1996) 221–235.
- [44] H.C. Lee, T. Chang, Characterization of binary polymer mixtures by simultaneous size exclusion chromatography and interaction chromatography, *Macromol.* 29 (22) (1996) 7294–7296.
- [45] H. Pasch, K. Rode, Chromatographic investigations of macromolecules in the critical range of liquid chromatography, 10. Polymer blend analysis using a viscosity detector, *Macromol. Chem. Phys.* 197 (9) (1996) 2691–2701.
- [46] H. Pasch, K. Rode, N. Chaumien, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 9. Separation of methacrylate-based polymer blends, *Polymer* 37 (18) (1996) 4079–4083.
- [47] H. Pasch, A. Deffieux, I. Henze, M. Schappacher, L. Rique-Lurbet, Analysis of macrocyclic polystyrenes. 1. Liquid chromatographic investigations, *Macromol.* 29 (27) (1996) 8776–8782.
- [48] H. Philipsen, B. Klumperman, A. van Herk, A. German, Critical retention behaviour of polymers a study on the influence of some practical parameters, *J. Chromatogr. A* 727 (1) (1996) 13–25.
- [49] D. Berek, M. Janco, K. Hatada, T. Kitayama, N. Fujimoto, Separation of poly(methyl methacrylate)s according to their tacticity II. Chromatographic investigations of poly(methyl methacrylate)s with different tacticity at the critical adsorption point, *Polym. J.* 29 (12) (1997) 1029–1033.
- [50] T. Chang, H.C. Lee, W. Lee, High performance liquid chromatography characterization of macromolecules, *Macromol. Symp.* 118 (1) (1997) 261–265.
- [51] G. Czichocki, R. Heger, W.A. Goedel, H. Much, Liquid chromatography of polyisoprenes: determination of critical composition using non protic binary and acidic ternary solvent mixtures, *J. Chromatogr. A* 791 (1–2) (1997) 350–356.
- [52] M. Janco, D. Berek, A. Önen, C.H. Fischer, Y. Yagci, W. Schnabel, Separation of block copolymers from parent homopolymers by means of liquid chromatography at the critical adsorption point, *Polym. Bull.* 38 (6) (1997) 681–688.

- [53] I. Souvignet, S.V. Olesik, Liquid chromatography at the critical condition using enhanced-fluidity liquid mobile phases, *Anal. Chem.* 69 (1) (1997) 66–71.
- [54] O. Wachsen, K. Reichert, R. Krüger, H. Much, G. Schulz, Thermal decomposition of biodegradable polyesters—III. Studies on the mechanisms of thermal degradation of oligo-l-lactide using SEC, LACCC and MALDI-TOF-MS, *Polym. Degrad. Stab.* 55 (2) (1997) 225–231.
- [55] D. Berek, M. Janco, G.R. Meira, Liquid chromatography of macromolecules at the critical adsorption point. II. Role of column packing: Bare silica gel, *J. Polym. Sci. Part A: Polym. Chem.* 36 (9) (1998) 1363–1371.
- [56] D. Berek, Liquid chromatography of macromolecules under limiting conditions of desorption. 1. Principles of the method, *Macromol.* 31 (24) (1998) 8517–8521.
- [57] D. Braun, E. Esser, H. Rasch, Chromatographic investigations of macromolecules in the critical range of liquid chromatography. XII. Analysis of block copolymers of styrene and butadiene, *Int. J. Polym. Anal. Charact.* 4 (6) (1998) 501–516.
- [58] A. Gorbunov, L. Ya Solov'yova, A. Skvortsov, General mechanism of the chromatography of macromolecules: Experimental test of the universal nature of adsorption effects, *Polymer* 39 (3) (1998) 697–702.
- [59] R. Murgasova, I. Capek, E. Lathova, D. Berek, S. Florian, Application of a new chromatographic method, LC CAP, in the characterization of polystyrene-graft-poly(ethylene oxide) copolymers, *Eur. Polym. J.* 34 (5–6) (1998) 659–663.
- [60] H. Pasch, K. Rode, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 11. Polymer blend separation using a reversed stationary phase, *Polymer* 39 (25) (1998) 6377–6383.
- [61] A. Skvortsov, A. Gorbunov, D. Berek, B. Trathnigg, Liquid chromatography of macromolecules at the critical adsorption point: behaviour of a polymer chain inside pores, *Polymer* 39 (2) (1998) 423–429.
- [62] H. Yun, S.V. Olesik, E.H. Marti, Improvements in polymer characterization by size-exclusion chromatography and liquid chromatography at the critical condition by using enhanced-fluidity liquid mobile phases with packed capillary columns, *Anal. Chem.* 70 (15) (1998) 3298–3303.
- [63] J. Adrian, D. Braun, K. Rode, H. Pasch, New analytical methods for epoxy resins, 1. Liquid chromatography at the critical point of adsorption and MALDI-TOF mass spectrometry, *Die Angew. Makromol. Chem.* 267 (1) (1999) 73–81.
- [64] J. Adrian, D. Braun, H. Pasch, New analytical methods for epoxy resins, 2. Two-dimensional chromatography, *Die Angew. Makromol. Chem.* 267 (1) (1999) 82–88.
- [65] K. Baran, S. Laugier, H. Cramail, Investigation of molecular dimensions of polystyrene as a function of the solvent composition: application to liquid chromatography at the exclusion-adsorption transition point, *Macromol. Chem. Phys.* 200 (9) (1999) 2074–2079.
- [66] Y. Brun, The mechanism of copolymer retention in interactive polymer chromatography. II. Gradient separation, *J. Liq. Chromatogr. Relat. Technol.* 22 (20) (1999) 3067–3090.
- [67] T. Chang, H.C. Lee, W. Lee, S. Park, C. Ko, Polymer characterization by temperature gradient interaction chromatography, *Macromol. Chem. Phys.* 200 (10) (1999) 2188–2204.
- [68] K.E. Esser, D. Braun, H. Pasch, Chromatographic investigations of macromolecules in the critical range of liquid chromatography, XIII. Separation of blends of styrene-butadiene rubber and butyl rubber, *Die Angew. Makromol. Chem.* 271 (1) (1999) 61–67.
- [69] R.-P. Krüger, H. Much, G. Schulz, E. Rikowski, Zur charakterisierung von si-polymeren durch flüssigchromatographische verfahren in kopplung mit MALDI-TOF-MS, *Monatsh. Chem. - Chem. Mon.* 130 (1) (1999) 163–174.
- [70] H. Lee, W. Lee, T. Chang, S. Choi, D. Lee, H. Ji, W.K. Nonidez, J.W. Mays, Characterization of poly(ethylene oxide)-block-poly(l-lactide) by HPLC and MALDI-TOF mass spectrometry, *Macromol.* 32 (12) (1999) 4143–4146.
- [71] J. Adrian, E. Esser, G. Hellmann, H. Pasch, Two-dimensional chromatography of complex polymers part 1. Analysis of a graft copolymer by two-dimensional chromatography with on-line FTIR detection, *Polymer* 41 (7) (2000) 2439–2449.
- [72] J. Falkenhagen, H. Much, W. Stauff, A.H.E. Müller, Characterization of block copolymers by liquid adsorption chromatography at critical conditions. 1. Diblock copolymers, *Macromol.* 33 (10) (2000) 3687–3693.
- [73] J. Falkenhagen, J.F. Friedrich, G. Schulz, R.-P. Krüger, H. Much, S. Weidner, Liquid adsorption Chromatography near Critical conditions of adsorption coupled with matrix-assisted laser desorption/ionization mass spectrometry, *Int. J. Polym. Anal. Charact.* 5 (4–6) (2000) 549–562.
- [74] M. Janco, T. Hirano, T. Kitayama, K. Hatada, D. Berek, Discrimination of poly(ethyl methacrylate)s according to their molar mass and tacticity by coupling size exclusion chromatography and liquid chromatography at the critical adsorption point, *Macromol.* 33 (5) (2000) 1710–1715.
- [75] T. Kitayama, M. Janco, K. Ute, R. Niimi, K. Hatada, D. Berek, Analysis of poly(ethyl methacrylate)s by on-line hyphenation of liquid chromatography at the critical adsorption point and nuclear magnetic resonance spectroscopy, *Anal. Chem.* 72 (7) (2000) 1518–1522.
- [76] H.C. Lee, H. Lee, W. Lee, T. Chang, J. Roovers, Fractionation of cyclic polystyrene from linear precursor by HPLC at the chromatographic critical condition, *Macromol.* 33 (22) (2000) 8119–8121.
- [77] B. Lepoittevin, M.-A. Dourges, M. Masure, P. Hemery, K. Baran, H. Cramail, Synthesis and characterization of ring-shaped polystyrenes, *Macromol.* 33 (22) (2000) 8218–8224.
- [78] K. Baran, S. Laugier, H. Cramail, Liquid chromatography of polymers at the exclusion — Adsorption transition point: Physicochemical interpretation, *Int. J. Polym. Anal. Charact.* 6 (1–2) (2001) 123–145.
- [79] K. Baran, S. Laugier, H. Cramail, Fractionation of functional polystyrenes, poly(ethylene oxide)s and poly(styrene)-b-poly(ethylene oxide) by liquid chromatography at the exclusion-adsorption transition point, *J. Chromatogr. B: Biomed. Sci. Appl.* 753 (1) (2001) 139–149.
- [80] I. Bruheim, P. Molander, M. Theodorsen, E. Ommundsen, E. Lundanes, T. Greibrokk, Temperature-programmed packed-capillary liquid chromatographic separation of polystyrenes, *Chromatogr.* 53 (S1) (2001) S266–S270.
- [81] D. Cho, S. Park, K. Kwon, T. Chang, J. Roovers, Structural characterization of ring polystyrene by liquid chromatography at the critical condition and MALDI-TOF mass spectrometry, *Macromol.* 34 (21) (2001) 7570–7572.
- [82] C. Keil, E. Esser, H. Pasch, Matrix-assisted laser desorption/ionization mass spectrometry of synthetic polymers, 5. Analysis of poly(propylene oxide)s by coupled liquid chromatography at the critical point of adsorption and MALDI-TOF mass spectrometry, *Macromol. Mater. Eng.* 286 (3) (2001) 161–167.
- [83] W. Lee, S. Park, T. Chang, Liquid chromatography at the critical condition for polyisoprene using a single solvent, *Anal. Chem.* 73 (16) (2001) 3884–3889.
- [84] W. Lee, D. Cho, T. Chang, K.J. Hanley, T.P. Lodge, Characterization of polystyrene-b-polyisoprene diblock copolymers by liquid chromatography at the chromatographic critical condition, *Macromol.* 34 (7) (2001) 2353–2358.
- [85] H. Lee, T. Chang, D. Lee, M.S. Shim, H. Ji, W.K. Nonidez, J.W. Mays, Characterization of poly(l-lactide)-block-poly-(ethylene oxide)-block-poly(l-lactide) triblock copolymer by liquid chromatography at the critical condition and by MALDI-TOF mass spectrometry, *Anal. Chem.* 73 (8) (2001) 1726–1732.
- [86] B. Lepoittevin, X. Perrot, M. Masure, P. Hemery, New route to synthesis of cyclic polystyrenes using controlled free radical polymerization, *Macromol.* 34 (3) (2001) 425–429.
- [87] B. Lepoittevin, P. Hemery, Synthesis of multicyclic and grafted polystyrenes, *J. Polym. Sci. Part A: Polym. Chem.* 39 (16) (2001) 2723–2730.
- [88] H. Pasch, E. Esser, C. Kloninger, H. Iatrou, N. Hadjichristidis, Chromatographic investigations of macromolecules in the critical range of liquid chromatography, 14. Analysis of miktoarm star (μ -star) polymers, *Macromol. Chem. Phys.* 202 (8) (2001) 1424–1429.
- [89] A. Siewing, J. Schierholz, D. Braun, G. Hellmann, H. Pasch, Two-dimensional chromatography of complex polymers, 2. Analysis of the grafting reaction of methyl methacrylate onto EPDM, *Macromol. Chem. Phys.* 202 (14) (2001) 2890–2894.
- [90] T. Macko, D. Hunkeler, D. Berek, Liquid chromatography of synthetic polymers under critical conditions. the case of single eluents and the role of θ conditions, *Macromol.* 35 (5) (2002) 1797–1804.
- [91] S. Phillips, S.V. Olesik, Fundamental studies of liquid chromatography at the critical condition using enhanced-fluidity liquids, *Anal. Chem.* 74 (4) (2002) 799–808.
- [92] T. Biela, A. Duda, K. Rode, H. Pasch, Characterization of star-shaped poly(l-lactide)s by liquid chromatography at critical conditions, *Polymer* 44 (6) (2003) 1851–1860.
- [93] S.M. Graef, A.J.P. Van Zyl, R.D. Sanderson, B. Klumperman, H. Pasch, Use of gradient, critical, and two-dimensional chromatography in the analysis of styrene- and methyl methacrylate-grafted epoxidized natural rubber, *J. Appl. Polym. Sci.* 88 (10) (2003) 2530–2538.
- [94] I. Park, S. Park, D. Cho, T. Chang, E. Kim, K. Lee, Y.J. Kim, Effect of block copolymer chain architecture on chromatographic retention, *Macromol.* 36 (22) (2003) 8539–8543.
- [95] S.L. Phillips, L. Ding, M. Stegemiller, S.V. Olesik, Development of water-based liquid chromatography at the critical condition, *Anal. Chem.* 75 (20) (2003) 5539–5543.
- [96] S.L. Phillips, S.V. Olesik, Initial characterization of humic acids using liquid chromatography at the critical condition followed by size-exclusion chromatography and electrospray ionization mass spectrometry, *Anal. Chem.* 75 (20) (2003) 5544–5553.
- [97] C. Rappel, B. Trathnigg, A. Gorbunov, Liquid chromatography of polyethylene glycol mono- and diesters: functional macromolecules or block copolymers? *J. Chromatogr. A* 984 (1) (2003) 29–43.
- [98] K. Im, S. Park, D. Cho, T. Chang, K. Lee, N. Choi, HPLC and MALDI-TOF MS analysis of highly branched polystyrene: Resolution enhancement by branching, *Anal. Chem.* 76 (9) (2004) 2638–2642.
- [99] H.J. Philipsen, Determination of chemical composition distributions in synthetic polymers, *J. Chromatogr. A* 1037 (1–2) (2004) 329–350.
- [100] X. Jiang, P.J. Schoenmakers, X. Lou, V. Lima, J.L. van Dongen, J. Brokken-Zijp, Separation and characterization of functional poly (n-butyl acrylate) by critical liquid chromatography, *J. Chromatogr. A* 1055 (1–2) (2004) 123–133.
- [101] M. Adler, F. Rittig, S. Becker, H. Pasch, Multidimensional chromatographic and hyphenated techniques for hydrophilic copolymers, *Macromol. Chem. Phys.* 206 (22) (2005) 2269–2277.

- [102] K.A. Alberty, T.E. Hogen-Esch, S. Carloti, Synthesis and characterization of macrocyclic vinyl-aromatic polymers, *Macromol. Chem. Phys.* 206 (10) (2005) 1035–1042.
- [103] M.A. Bashir, A. Brüll, W. Radke, Fast determination of critical eluent composition for polymers by gradient chromatography, *Polymer* 46 (10) (2005) 3223–3229.
- [104] D. Cho, K. Masuoka, K. Koguchi, T. Asari, D. Kawaguchi, A. Takano, Y. Matsushita, Preparation and characterization of cyclic polystyrenes, *Polym. J.* 37 (7) (2005) 506–511.
- [105] L. Coulier, E. Kaal, T. Hankemeier, Comprehensive two-dimensional liquid chromatography and hyphenated liquid chromatography to study the degradation of poly(bisphenol A)carbonate, *J. Chromatogr. A* 1070 (1–2) (2005) 79–87.
- [106] L.-C. Heinz, S. Graef, T. Macko, R. Brüll, S. Balk, H. Keul, H. Pasch, Development of high-temperature separation techniques for the chemical composition analysis of ethylene-methyl methacrylate block copolymers, *E-Polym.* 5 (1) (2005).
- [107] X. Jiang, A. van der Horst, V. Lima, P.J. Schoenmakers, Comprehensive two-dimensional liquid chromatography for the characterization of functional acrylate polymers, *J. Chromatogr. A* 1076 (1–2) (2005) 51–61.
- [108] V. Lima, X. Jiang, J. Brokken-Zijp, P.J. Schoenmakers, B. Klumperman, R. Van Der Linde, Synthesis and characterization of telechelic polymethacrylates via RAFT polymerization, *J. Polym. Sci. Part A: Polym. Chem.* 43 (5) (2005) 959–973.
- [109] K. Min, H. Gao, K. Matyjaszewski, Preparation of homopolymers and block copolymers in miniemulsion by ATRP using activators generated by electron transfer (AGET), *J. Am. Chem. Soc.* 127 (11) (2005) 3825–3830.
- [110] H. Pasch, M. Adler, D. Knecht, F. Rittig, R. Lange, New chromatographic and hyphenated techniques for hydrophilic copolymers, *Macromol. Symp.* 231 (1) (2005) 166–177.
- [111] W. Radke, K. Rode, A.V. Gorshkov, T. Biela, Chromatographic behavior of functionalized star-shaped poly(lactide)s under critical conditions of adsorption. Comparison of theory and experiment, *Polymer* 46 (15) (2005) 5456–5465.
- [112] B. Trathnigg, Characterization of amphiphilic polymers: Independent analysis of blocks in poloxamers by liquid chromatography under critical conditions, *Polymer* 46 (22) (2005) 9211–9223.
- [113] B. Trathnigg, C. Rappel, S. Fraydl, A. Gorbunov, Liquid chromatography of polyoxyethylenes under critical conditions: A thermodynamic study, *J. Chromatogr. A* 1085 (2) (2005) 253–261.
- [114] M.A. Bashir, W. Radke, Comparison of retention models for polymers, *J. Chromatogr. A* 1131 (1–2) (2006) 130–141.
- [115] L. Coulier, E. Kaal, T. Hankemeier, Towards early detection of the hydrolytic degradation of poly(bisphenol A)carbonate by hyphenated liquid chromatography and comprehensive two-dimensional liquid chromatography, *Polym. Degrad. Stab.* 91 (2) (2006) 271–279.
- [116] L. Coulier, E. Kaal, T. Hankemeier, Hyphenation of infrared spectroscopy to liquid chromatography for qualitative and quantitative polymer analysis: Degradation of poly(bisphenol A)carbonate, *J. Chromatogr. A* 1130 (1) (2006) 34–42.
- [117] H. Gao, K. Min, K. Matyjaszewski, Characterization of linear and 3-arm star block copolymers by liquid chromatography at critical conditions, *Macromol. Chem. Phys.* 207 (19) (2006) 1709–1717.
- [118] K. Im, Y. Kim, T. Chang, K. Lee, N. Choi, Separation of branched polystyrene by comprehensive two-dimensional liquid chromatography, *J. Chromatogr. A* 1103 (2) (2006) 235–242.
- [119] D. Knecht, F. Rittig, R.F. Lange, H. Pasch, Multidimensional chromatographic techniques for hydrophilic copolymers, *J. Chromatogr. A* 1130 (1) (2006) 43–53.
- [120] B. Trathnigg, A. Gorbunov, Characterization of EO-PO block copolymers by liquid chromatography under critical conditions, *Macromol. Symp.* 237 (1) (2006) 18–27.
- [121] A. Albrecht, R. Brüll, T. Macko, H. Pasch, Separation of ethylene-vinyl acetate copolymers by high-temperature gradient liquid chromatography, *Macromol.* 40 (15) (2007) 5545–5551.
- [122] H. Gao, K. Matyjaszewski, Arm-first method as a simple and general method for synthesis of miktoarm star copolymers, *J. Am. Chem. Soc.* 129 (38) (2007) 11828–11834.
- [123] M.d. Geus, Enzymatic Catalysis in the Synthesis of New Polymer Architectures and Materials (PhD thesis), Technische Universiteit Eindhoven, 2007.
- [124] W. Hiller, P. Sinha, H. Pasch, On-line HPLC-NMR of PS-b-PMMA and blends of PS and PMMA: LCCC-NMR at critical conditions of PS, *Macromol. Chem. Phys.* 208 (18) (2007) 1965–1978.
- [125] M. Jacquin, P. Muller, G. Lizarraga, C. Bauer, H. Cottet, O. Théodoly, Characterization of amphiphilic diblock copolymers synthesized by MADIX polymerization process, *Macromol.* 40 (8) (2007) 2672–2682.
- [126] M. Jacquin, P. Muller, R. Talingting-Pabalan, H. Cottet, J. Berret, T. Futterer, O. Théodoly, Chemical analysis and aqueous solution properties of charged amphiphilic block copolymers PBA-b-PAA synthesized by MADIX®, *J. Colloid Interface Sci.* 316 (2) (2007) 897–911.
- [127] M.I. Malik, B. Trathnigg, C.O. Kappe, Selectivity of PEO-block-PPO diblock copolymers in the microwave-accelerated, anionic ring-opening polymerization of propylene oxide with PEG as initiator, *Macromol. Chem. Phys.* 208 (23) (2007) 2510–2524.
- [128] C. Petit, B. Luneau, E. Beaudoin, D. Gimes, D. Bertin, Liquid chromatography at the critical conditions in pure eluent: An efficient tool for the characterization of functional polystyrenes, *J. Chromatogr. A* 1163 (1–2) (2007) 128–137.
- [129] A. Takano, Y. Kushida, K. Aoki, K. Masuoka, K. Hayashida, D. Cho, D. Kawaguchi, Y. Matsushita, HPLC characterization of cyclization reaction product obtained by end-to-end ring closure reaction of a telechelic polystyrene, *Macromol.* 40 (3) (2007) 679–681.
- [130] S. Trimpin, S.M. Weidner, J. Falkenhagen, C.N. McEwen, Fractionation and solvent-free MALDI-MS analysis of polymers using liquid adsorption chromatography at critical conditions in combination with a multisample on-target homogenization/transfer sample preparation method, *Anal. Chem.* 79 (19) (2007) 7565–7570.
- [131] S.M. Weidner, J. Falkenhagen, S. Maltsev, V. Sauerland, M. Rinken, A novel software tool for copolymer characterization by coupling of liquid chromatography with matrix-assisted laser desorption/ionization time-of-flight mass spectrometry, *Rapid Commun. Mass Spectrom.* 21 (16) (2007) 2750–2758.
- [132] S. Weidner, J. Falkenhagen, R.-P. Krueger, U. Just, Principle of two-dimensional characterization of copolymers, *Anal. Chem.* 79 (13) (2007) 4814–4819.
- [133] E. Beaudoin, A. Favier, C. Galindo, A. Lapp, C. Petit, D. Gimes, S. Marqu, D. Bertin, Reduced sample recovery in liquid chromatography at critical adsorption point of high molar mass polystyrene, *Eur. Polym. J.* 44 (2) (2008) 514–522.
- [134] J. Falkenhagen, S. Weidner, Determination of critical conditions of adsorption for chromatography of polymers, *Anal. Chem.* 81 (1) (2008) 282–287.
- [135] H. Gao, K. Matyjaszewski, Synthesis of low-polydispersity miktoarm star copolymers via a simple ‘Arm-First’ method: Macromonomers as arm precursors, *Macromol.* 41 (12) (2008) 4250–4257.
- [136] M.d. Geus, R. Peters, C.E. Koning, A. Heise, Insights into the initiation process of enzymatic ring-opening polymerization from monofunctional alcohols using liquid chromatography under critical conditions, *Biomacromolecules* 9 (2) (2008) 752–757.
- [137] M. Girod, T.N.T. Phan, L. Charles, On-line coupling of liquid chromatography at critical conditions with electrospray ionization tandem mass spectrometry for the characterization of a nitroxide-mediated poly(ethylene oxide)/polystyrene block copolymer, *Rapid Commun. Mass Spectrom.* 22 (23) (2008) 3767–3775.
- [138] A.S. Goldmann, D. Quémener, P.-E. Millard, T.P. Davis, M.H. Stenzel, C. Barner-Kowollik, A.H. Müller, Access to cyclic polystyrenes via a combination of reversible addition fragmentation chain transfer (RAFT) polymerization and click chemistry, *Polymer* 49 (9) (2008) 2274–2281.
- [139] K. Im, H.-W. Park, Y. Kim, S. Ahn, T. Chang, K. Lee, H.-J. Lee, J. Ziebarth, Y. Wang, Retention behavior of star-shaped polystyrene near the chromatographic critical condition, *Macromol.* 41 (9) (2008) 3375–3383.
- [140] J. Listak, W. Jakubowski, L. Mueller, A. Plichta, K. Matyjaszewski, M.R. Bockstaller, Effect of symmetry of molecular weight distribution in block copolymers on formation of ‘metastable’ morphologies, *Macromol.* 41 (15) (2008) 5919–5927.
- [141] M.I. Malik, B. Trathnigg, C.O. Kappe, Microwave assisted synthesis and characterization of end functionalized poly(propylene oxide) as model compounds, *Eur. Polym. J.* 44 (1) (2008) 144–154.
- [142] V. Mass, V. Bellas, H. Pasch, Two-dimensional chromatography of complex polymers, 7 – detailed study of polystyrene-block-polyisoprene diblock copolymers prepared by sequential anionic polymerization and coupling chemistry, *Macromol. Chem. Phys.* 209 (19) (2008) 2026–2039.
- [143] S. Park, D.Y. Ryu, J.K. Kim, M. Ree, T. Chang, Property of diblock copolymer having extremely narrow molecular weight distribution, *Polymer* 49 (8) (2008) 2170–2175.
- [144] H. Pasch, L.-C. Heinz, T. Macko, W. Hiller, High-temperature gradient HPLC and LC-NMR for the analysis of complex polyolefins, *Pure Appl. Chem.* 80 (8) (2008) 1747–1762.
- [145] A.I. Abdulahad, C.Y. Ryu, Liquid chromatography at the critical condition: Thermodynamic significance and influence of pore size, *J. Polym. Sci. Part B: Polym. Phys.* 47 (24) (2009) 2533–2540.
- [146] H. Ahmed, B. Trathnigg, C. Oliver Kappe, R. Saf, Characterization of poly(ethylene glycol)-b-poly(ϵ -caprolactone) by liquid chromatography under critical conditions: Influence of catalysts and reaction conditions on product composition, *Eur. Polym. J.* 45 (8) (2009) 2338–2347.
- [147] H. Ahmed, B. Trathnigg, Characterization of poly(ethylene glycol)-b-poly(ϵ -caprolactone) by two-dimensional liquid chromatography, *J. Sep. Sci.* 32 (9) (2009) 1390–1400.
- [148] C. Galindo, E. Beaudoin, D. Gimes, D. Bertin, Polybutadiene-graft-polystyrene copolymer: Grafting quantification by liquid chromatography at critical conditions using single UV detection, *J. Chromatogr. A* 1216 (47) (2009) 8386–8390.
- [149] M. Girod, E. Beaudoin, L. Charles, Effect of salt in the mobile phase on the critical conditions of poly(ethylene glycol) in liquid chromatography-mass spectrometry coupling, *Anal. Methods* 1 (2) (2009) 128.

- [150] M. Girod, T.N.T. Phan, L. Charles, Tuning block copolymer structural information by adjusting salt concentration in liquid chromatography at critical conditions coupled with electrospray tandem mass spectrometry, *Rapid Commun. Mass Spectrom.* 23 (10) (2009) 1476–1482.
- [151] J. Han, B.H. Jeon, C.Y. Ryu, J.J. Semler, Y.K. Jhon, J. Genzer, Discriminating among co-monomer sequence distributions in random copolymers using interaction chromatography, *Macromol. Rapid Commun.* 30 (18) (2009) 1543–1548.
- [152] W. Hiller, P. Sinha, H. Pasch, Online HPLC-NMR of PS-b-PMMA and blends of PS and PMMA, 2-LCCC-NMR at critical conditions of PMMA, *Macromol. Chem. Phys.* 210 (8) (2009) 605–613.
- [153] K. Im, H.-w. Park, S. Lee, T. Chang, Two-dimensional liquid chromatography analysis of synthetic polymers using fast size exclusion chromatography at high column temperature, *J. Chromatogr. A* 1216 (21) (2009) 4606–4610.
- [154] S. Julka, H. Cortes, R. Harfmann, B. Bell, A. Schweizer-Theobaldt, M. Pursch, L. Mondello, S. Maynard, D. West, Quantitative characterization of solid epoxy resins using comprehensive two dimensional liquid chromatography coupled with electrospray ionization-time of flight mass spectrometry, *Anal. Chem.* 81 (11) (2009) 4271–4279.
- [155] M.I. Malik, H. Ahmed, B. Trathnigg, Liquid chromatography under critical conditions: Practical applications in the analysis of amphiphilic polymers, *Anal. Bioanal. Chem.* 393 (6–7) (2009) 1797–1804.
- [156] M.I. Malik, B. Trathnigg, R. Saf, Characterization of ethylene oxide-propylene oxide block copolymers by combination of different chromatographic techniques and matrix-assisted laser desorption ionization time-of-flight mass spectroscopy, *J. Chromatogr. A* 1216 (38) (2009) 6627–6635.
- [157] M.I. Malik, B. Trathnigg, C.O. Kappe, Amphiphilic polymers based on higher alkylene oxides, *J. Chromatogr. A* 1216 (7) (2009) 1167–1173.
- [158] M.I. Malik, B. Trathnigg, C. Oliver Kappe, Microwave-assisted polymerization of higher alkylene oxides, *Eur. Polym. J.* 45 (3) (2009) 899–910.
- [159] M.I. Malik, B. Trathnigg, Full separation of oligomers in block copolymers of ethylene oxide and propylene oxide, *J. Sep. Sci.* 32 (11) (2009) 1771–1781.
- [160] A. Takano, Y. Kushida, Y. Ohta, K. Masuoka, Y. Matsushita, The second virial coefficients of highly-purified ring polystyrenes in cyclohexane, *Polymer* 50 (5) (2009) 1300–1303.
- [161] S. Abrar, B. Trathnigg, Characterization of polyoxyethylenes according to the number of hydroxy end groups by hydrophilic interaction chromatography at critical conditions for polyethylene glycol, *Anal. Bioanal. Chem.* 400 (8) (2010) 2531–2537.
- [162] H. Ahmed, B. Trathnigg, C.O. Kappe, R. Saf, Synthesis of poly(ϵ -caprolactone) diols and EO-CL block copolymers and their characterization by liquid chromatography and MALDI-TOF-MS, *Eur. Polym. J.* 46 (3) (2010) 494–505.
- [163] H. Ahmed, B. Trathnigg, Characterization of PEG in monomethyl ethers of PEG by 2-D LC, *J. Sep. Sci.* 33 (10) (2010) 1448–1453.
- [164] M. Al Samman, W. Radke, A. Khalyavina, A. Lederer, Retention behavior of linear, branched, and hyperbranched polyesters in interaction liquid chromatography, *Macromol.* 43 (7) (2010) 3215–3220.
- [165] W.K. Elhrari, P.E. Mallon, Two-dimensional chromatographic analysis of polystyrene-block-poly(methyl methacrylate) copolymers synthesized by selective oxidation of polystyrene-9-borabicyclo[3.3.1]nonane, *Macromol. Symp.* 298 (1) (2010) 174–181.
- [166] N. Fandrich, J. Falkenhagen, S.M. Weidner, B. Staal, A.F. Thünemann, A. Laschewsky, Characterization of new amphiphilic block copolymers of N-vinylpyrrolidone and vinyl acetate, 2-chromatographic separation and analysis by MALDI-TOF and FT-IR coupling, *Macromol. Chem. Phys.* 211 (15) (2010) 1678–1688.
- [167] W. Hiller, H. Pasch, P. Sinha, T. Wagner, J. Thiel, M. Wagner, K. Müllen, Coupling of NMR and liquid chromatography at critical conditions: A new tool for the block length and microstructure analysis of block copolymers, *Macromol.* 43 (11) (2010) 4853–4863.
- [168] M.I. Malik, B. Trathnigg, K. Bartl, R. Saf, Characterization of polyoxyalkylene block copolymers by combination of different chromatographic techniques and MALDI-TOF-MS, *Anal. Chim. Acta* 658 (2) (2010) 217–224.
- [169] C. Schmid, J. Falkenhagen, C. Barner-Kowollik, An efficient avenue to poly(styrene)-block-poly(ϵ -caprolactone) polymers via switching from RAFT to hydroxyl functionality: Synthesis and characterization, *J. Polym. Sci. Part A: Polym. Chem.* 49 (1) (2010) 1–10.
- [170] P. Sinha, W. Hiller, H. Pasch, Characterisation of blends of polyisoprene and polystyrene by on-line hyphenation of HPLC and 1H-NMR: LC-CC-NMR at critical conditions of both homopolymers, *J. Sep. Sci.* 33 (22) (2010) 3494–3500.
- [171] B. Trathnigg, M.I. Malik, N. Pircher, S. Hayden, Liquid chromatography at critical conditions in ternary mobile phases: Gradient elution along the critical line, *J. Sep. Sci.* 33 (14) (2010) 2052–2059.
- [172] B. Trathnigg, H. Ahmed, Separation of all oligomers in polyethylene glycols and their monomethyl ethers by one-dimensional liquid chromatography, *Anal. Bioanal. Chem.* 399 (4) (2010) 1535–1545.
- [173] B. Trathnigg, S. Abrar, Characterization of complex copolymers by two-dimensional liquid chromatography, *Procedia Chem.* 2 (1) (2010) 130–139.
- [174] M. van Hulst, A. van der Horst, W.T. Kok, P.J. Schoenmakers, Comprehensive 2-D chromatography of random and block methacrylate copolymers, *J. Sep. Sci.* 33 (10) (2010) 1414–1420.
- [175] E.H.H. Wong, M.H. Stenzel, T. Junkers, C. Barner-Kowollik, Spin capturing with ‘clickable’ nitrones: Generation of miktoarmed star polymers, *Macromol.* 43 (8) (2010) 3785–3793.
- [176] A. Baumgaertel, C. Weber, N. Fritz, G. Festag, E. Altuntaş, K. Kempe, R. Hoogenboom, U.S. Schubert, Characterization of poly(2-oxazoline) homo- and copolymers by liquid chromatography at critical conditions, *J. Chromatogr. A* 1218 (46) (2011) 8370–8378.
- [177] W. Hiller, P. Sinha, M. Hehn, H. Pasch, T. Hofe, Separation analysis of polyisoprenes regarding microstructure by online LCCC-NMR and SEC-NMR, *Macromol.* 44 (6) (2011) 1311–1318.
- [178] A.J. Inglis, C. Barner-Kowollik, Visualizing the efficiency of rapid modular block copolymer construction, *Polym. Chem.* 2 (1) (2011) 126–136.
- [179] M.I. Malik, P. Sinha, G.M. Bayley, P.E. Mallon, H. Pasch, Characterization of polydimethylsiloxane-block-polystyrene (PDMS-b-PS) copolymers by liquid chromatography at critical conditions, *Macromol. Chem. Phys.* 212 (12) (2011) 1221–1228.
- [180] C. Schmid, S. Weidner, J. Falkenhagen, C. Barner-Kowollik, In-depth LCCC-(GELC)-SEC characterization of ABA block copolymers generated by a mechanistic switch from RAFT to ROP, *Macromol.* 45 (1) (2011) 87–99.
- [181] A. Takano, Y. Ohta, K. Masuoka, K. Matsubara, T. Nakano, A. Hieno, M. Itakura, K. Takahashi, S. Kinugasa, D. Kawaguchi, Y. Takahashi, Y. Matsushita, Radii of gyration of ring-shaped polystyrenes with high purity in dilute solutions, *Macromol.* 45 (1) (2011) 369–373.
- [182] H. Barqawi, E. Oostas, B. Liu, J.-F. Carpentier, W.H. Binder, Multidimensional characterization of α,ω -telechelic poly(ϵ -caprolactone)s via online coupling of 2D chromatographic methods (LC/SEC) and ESI-TOF/MALDI-TOF-MS, *Macromol.* 45 (24) (2012) 9779–9790.
- [183] M.I. Malik, G.W. Harding, M.E. Grabowsky, H. Pasch, Two-dimensional liquid chromatography of polystyrene-polyethylene oxide block copolymers, *J. Chromatogr. A* 1244 (2012) 77–87.
- [184] M.I. Malik, G.W. Harding, H. Pasch, Two-dimensional liquid chromatography of PDMS-PS block copolymers, *Anal. Bioanal. Chem.* 403 (2) (2012) 601–611.
- [185] Y. Ohta, M. Nakamura, Y. Matsushita, A. Takano, Synthesis, separation and characterization of knotted ring polymers, *Polymer* 53 (2) (2012) 466–470.
- [186] M. Rollet, D. Glé, T.N.T. Phan, Y. Guilleaume, D. Bertin, D. Gimes, Characterization of functional poly(ethylene oxide)s and their corresponding polystyrene block copolymers by liquid chromatography under critical conditions in organic solvents, *Macromol.* 45 (17) (2012) 7171–7178.
- [187] C. Schmid, J. Falkenhagen, T.F. Beskers, L.-T.T. Nguyen, M. Wilhelm, F.E. Du Prez, C. Barner-Kowollik, Multi-block polyurethanes via RAFT end-group switching and their characterization by advanced hyphenated techniques, *Macromol.* 45 (16) (2012) 6353–6362.
- [188] P. Sinha, G.W. Harding, K. Maiko, W. Hiller, H. Pasch, Comprehensive two-dimensional liquid chromatography for the separation of protonated and deuterated polystyrene, *J. Chromatogr. A* 1265 (2012) 95–104.
- [189] P. Sinha, W. Hiller, V. Bellas, H. Pasch, Analysis of polystyrene-b-polyisoprene copolymers by coupling of liquid chromatography at critical conditions to NMR at critical conditions of polystyrene and polyisoprene, *J. Sep. Sci.* 35 (14) (2012) 1731–1740.
- [190] P. Sinha, M. Grabowsky, M.I. Malik, G.W. Harding, H. Pasch, Characterization of polystyrene-block-polyethylene oxide diblock copolymers and blends of homopolymers by liquid chromatography at critical conditions (LCCC), *Macromol. Symp.* 313–314 (1) (2012) 162–169.
- [191] S.M. Weidner, J. Falkenhagen, I. Bressler, Copolymer composition determined by LC-MALDI-TOF MS coupling and ‘MassChrom2D’ data analysis, *Macromol. Chem. Phys.* 213 (22) (2012) 2404–2411.
- [192] A.-M. Zorn, C. Barner-Kowollik, Transformation of macromonomers into ring-opening polymerization macroinitiators: A detailed initiation efficiency study, *J. Polym. Sci. Part A: Polym. Chem.* 50 (12) (2012) 2366–2377.
- [193] H. Barqawi, M. Schulz, A. Olubummo, V. Saurland, W.H. Binder, 2D-LC/SEC-(MALDI-TOF)-MS characterization of symmetric and nonsymmetric biocompatible PEOm-PIB-PEOn block copolymers, *Macromol.* 46 (19) (2013) 7638–7649.
- [194] Y. Doi, Y. Ohta, M. Nakamura, A. Takano, Y. Takahashi, Y. Matsushita, Precise synthesis and characterization of tadpole-shaped polystyrenes with high purity, *Macromol.* 46 (3) (2013) 1075–1081.
- [195] D.J. Keddie, C. Guerrero-Sanchez, G. Moad, The reactivity of N-vinylcarbazole in RAFT polymerization: trithiocarbonates deliver optimal control for the synthesis of homopolymers and block copolymers, *Polym. Chem.* 4 (12) (2013) 3591.
- [196] M.I. Malik, H. Ahmed, B. Trathnigg, Separation of telechelic oligomers according to architecture by liquid chromatography, *J. Chromatogr. A* 1314 (2013) 180–187.
- [197] D. Mekap, T. Macko, R. Brüll, R. Cong, A. deGroot, A. Parrott, P. Cools, W. Yau, Liquid chromatography at critical conditions of polyethylene, *Polymer* 54 (21) (2013) 5518–5524.

- [198] M. Mlynek, W. Radke, Critical chromatography in ternary solvents, *J. Chromatogr. A* 1284 (2013) 112–117.
- [199] A.S. Kasegaonkar, H. Barqawi, W.H. Binder, Synthesis and crystallization of star-shaped photocleavable poly(ϵ -caprolactone)s, *J. Polym. Sci. Part A: Polym. Chem.* 53 (5) (2014) 642–649.
- [200] M.I. Malik, T. Mahboob, S. Ahmed, Characterization of poly(2-vinylpyridine)-block-poly(methyl methacrylate) copolymers and blends of their homopolymers by liquid chromatography at critical conditions, *Anal. Bioanal. Chem.* 406 (25) (2014) 6311–6317.
- [201] M.P.F. Pepels, P. Soullé, R. Peters, R. Duchateau, Theoretical and experimental approach to accurately predict the complex molecular weight distribution in the polymerization of stainless cyclic esters, *Macromol.* 47 (16) (2014) 5542–5550.
- [202] P. Sinha, W. Hiller, H. Pasch, HPLC-1H-NMR characterization of polystyrene-block-polyisoprene copolymers: LCCC-1H-NMR using a single mobile phase, *Macromol. Symp.* 337 (1) (2014) 44–50.
- [203] S. Banerjee, P.N. Shah, Y. Jeong, T. Chang, K. Seethamraju, R. Faust, Structural characterization of telechelic polyisobutylene diol, *J. Chromatogr. A* 1376 (2015) 98–104.
- [204] J.L. Fischer, C.A. Lutomski, T.J. El-Baba, B.N. Siriwardena-Mahanama, S.M. Weidner, J. Falkenhagen, M.J. Allen, S. Trimpin, Matrix-assisted ionization-ion mobility spectrometry-mass spectrometry: Selective analysis of a europium-PEG complex in a crude mixture, *J. Am. Soc. Mass Spectrom.* 26 (12) (2015) 2086–2095.
- [205] M. Hehn, P. Sinha, H. Pasch, W. Hiller, Onflow liquid chromatography at critical conditions coupled to 1H and 2H nuclear magnetic resonance as powerful tools for the separation of poly(methylmethacrylate) according to isotopic composition, *J. Chromatogr. A* 1387 (2015) 69–74.
- [206] T.G. Wright, H. Pasch, A simple route to deuterated polystyrene block copolymers by reverse iodine transfer polymerisation, *Polym. Chem.* 6 (17) (2015) 3236–3244.
- [207] C. Appiah, K.R. Siefertmann, M. Jorewitz, H. Barqawi, W.H. Binder, Synthesis and characterization of new photoswitchable azobenzene-containing poly(ϵ -caprolactones), *RSC Adv.* 6 (8) (2016) 6358–6367.
- [208] S. Bhati, T. Macko, R. Brüll, Estimation of critical conditions of polymers based on monitoring the polymer recovery, *J. Chromatogr. A* 1451 (2016) 91–96.
- [209] L. Gao, J. Oh, T. Chang, D. Chen, X. Li, X. Yang, Y. Tu, X. Zhu, C.Y. Li, A nearly quantitative synthetic approach towards monocyclic polystyrenes and the solvent, concentration and molecular weight effect on cyclic yield, *Polymer* 101 (2016) 379–387.
- [210] M.I. Malik, S. Lee, T. Chang, Comprehensive two-dimensional liquid chromatographic analysis of poloxamers, *J. Chromatogr. A* 1442 (2016) 33–41.
- [211] M.I. Malik, M. Irfan, A. Khan, S. Rahim, R. Abdul-Karim, J. Hashim, Alkylene oxide polymerizations: identification of side reactions and by-products, *J. Polym. Res.* 23 (12) (2016).
- [212] Y.-Z. Wei, R.-X. Zhuo, X.-L. Jiang, Separation of polyethylene glycols and amino-terminated polyethylene glycols by high-performance liquid chromatography under near critical conditions, *J. Chromatogr. A* 1447 (2016) 122–128.
- [213] Y.-Z. Wei, Y.-F. Chu, E. Uliyanchenko, P.J. Schoenmakers, R.-X. Zhuo, X.-L. Jiang, Separation and characterization of benzaldehyde-functional polyethylene glycols by liquid chromatography under critical conditions, *Polym. Chem.* 7 (48) (2016) 7506–7513.
- [214] R. Abdul-Karim, S.G. Musharraf, M.I. Malik, Synthesis and characterization of novel biodegradable di- and tri-block copolymers based on ethylene carbonate polymer as hydrophobic segment, *J. Polym. Sci. Part A: Polym. Chem.* 55 (11) (2017) 1887–1893.
- [215] J. Ahn, T. Chang, X. Wang, Z. Limpouchová, K. Procházka, Influence of the chain architecture and the presence of end-groups or branching units chemically different from repeating structural units on the critical adsorption point in liquid chromatography, *Macromol.* 50 (21) (2017) 8720–8730.
- [216] N. Apel, E. Uliyanchenko, S. Moyses, S. Rommens, C. Wold, T. Macko, K. Rode, R. Brüll, Selective chromatographic separation of polycarbonate according to hydroxyl end-groups using a porous graphitic carbon column, *J. Chromatogr. A* 1488 (2017) 77–84.
- [217] Y. Jeong, Y. Jin, T. Chang, F. Uhlik, J. Roovers, Intrinsic viscosity of cyclic polystyrene, *Macromol.* 50 (19) (2017) 7770–7776.
- [218] S. Lee, H. Lee, T. Chang, A. Hirao, Synthesis and characterization of an exact polystyrene-graft-polyisoprene: A failure of size exclusion chromatography analysis, *Macromol.* 50 (7) (2017) 2768–2776.
- [219] M.K. Tufail, R. Abdul-Karim, S. Rahim, S.G. Musharraf, M.I. Malik, Analysis of individual block length of amphiphilic di- and tri-block copolymers containing poly(ethylene oxide) and poly(methyl methacrylate), *RSC Adv.* 7 (66) (2017) 41693–41704.
- [220] S. Abrar, B. Trathnigg, S. Javed, S. Kiran, T. Gulzar, Characterization of tween® surfactants by MALDI TOF-MS and high performance liquid chromatography in a ternary mobile phase, *Tenside Surfactants Deterg.* 55 (6) (2018) 447–454.
- [221] L. Gao, J. Oh, Y. Tu, T. Chang, Preparation of low molecular weight cyclic polystyrenes with high purity via liquid chromatography at the critical condition, *Polymer* 135 (2018) 279–284.
- [222] S. Lee, H. Choi, T. Chang, B. Staal, Two-dimensional liquid chromatography analysis of polystyrene/polybutadiene block copolymers, *Anal. Chem.* 90 (10) (2018) 6259–6266.
- [223] R. Abdul-Karim, S.G. Musharraf, M.I. Malik, Poly(propylene ether carbonate)-based di- and tri-block copolymers: Synthesis and chromatographic characterization, *Macromol. Res.* 27 (9) (2019) 911–918.
- [224] R. Khatoun, S. Rahim, R. Abdul-Karim, S.G. Musharraf, M.I. Malik, Characterization of polystyrene-block-poly(2-vinyl pyridine) copolymers and blends of their homopolymers by liquid chromatography at critical conditions, *Macromol.* 52 (20) (2019) 7688–7695.
- [225] M.I. Malik, Critical parameters of liquid chromatography at critical conditions in context of poloxamers: Pore diameter, mobile phase composition, temperature and gradients, *J. Chromatogr. A* 1609 (2020) 460440.
- [226] T. Ahmad, S.S. Durrani, F.J. Iftikhar, S.G. Musharraf, M.I. Malik, Concurrent ring-opening and atom transfer radical polymerization for synthesis of block copolymers, and their comprehensive chromatographic characterization, *Eur. Polym. J.* 142 (2021) 110161.
- [227] K. Molnar, C.A. Helfer, G. Kaszas, E. Krisch, D. Chen, G.B. McKenna, J.A. Kornfield, J.E. Puskas, Liquid chromatography at critical conditions (LCCC): Capabilities and limitations for polymer analysis, *J. Mol. Liq.* 322 (2021) 114956.
- [228] S.K. Panda, T.A. Al-Qunaysi, M. Al Taha, D.J. Kearney, Size exclusion chromatography of petroleum samples using amino-bonded silica phase, *Fuel* 302 (2021) 120914.
- [229] J. Yang, L. Chen, M. Zhu, M.W. Ishaq, S. Chen, L. Li, Investigation of the multimer cyclization effect during click step-growth polymerization of AB-type macromonomers, *Macromol.* 55 (15) (2022) 6830–6840.
- [230] B. Zdovc, H. Li, J. Zhao, D. Pahovnik, E. Žagar, Influence of microstructure on the elution behavior of gradient copolymers in different modes of liquid interaction chromatography, *Anal. Chem.* 94 (22) (2022) 7844–7852.
- [231] S. Deshmukh, T. Macko, J.-H. Arndt, F. Malz, R. Bernardo, S. Niessen, G. van Doremale, R. Brüll, Critical conditions for liquid chromatography of statistical polyolefins: Evaluation of diene distribution in EPDM terpolymers, *Anal. Chim. Acta* 1246 (2023) 340856.
- [232] M.I. Malik, Novel epoxy-terminated macromonomers and their polymerization for synthesis of bottle-brush type amphiphilic block copolymers, *RSC Adv.* 13 (40) (2023) 28288–28298.